DOMINATING A PROPER MODIFICATION BY BLOWUPS IN CLOSED POINTS OVER A FINITE SET OF REGULAR SURFACE POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to an admissible blowing up supported on the finite set, the principalization along closed points above that set, and the application of the universal property.

Editorial note. The label lemma-dominate-by-blowing-up-in-points names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-dominate-by-blowing-up-in-points –start 1145 –end 1179 –local-id 28112 returned RESULT=OK (L28112 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Domination of a modification by an admissible blowup. More on Flatness, the lemma dominating a modification by an admissible blowup. Replaces the given proper morphism by a blowing up in an ideal whose quotient is supported on the finite set of points.
- Making an ideal principal by a sequence of blowups in closed points. Same chapter, the lemma producing a sequence of blowups in closed points above the given set after which the ideal becomes invertible. Provides the chain of blowups along which the ideal sheaf of the admissible blowup becomes invertible.
- The universal property of blowing up. Divisors, the universal property of the blowing up in an ideal. Produces the factorization of the composition through the given proper morphism once the ideal becomes invertible.

Source reference. Original lemma 0806: sources/stacks-chow/tags.txt, source lines 10171–10171. Prerequisite pointer invoked at proof line(s) 1177; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 081T: sources/stacks-chow/tags.txt, source lines 10228–10228. Prerequisite pointer invoked at proof line(s) 1161; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AHH: sources/stacks-resolve/pinned-original-source.tex, source lines 1062–1062. Prerequisite pointer invoked at proof line(s) 1167; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a Noetherian scheme. Let $T \subset X$ be a finite set of closed points x such that $\mathcal{O}_{X,x}$ is a regular local ring of dimension 2. Let $f : Y \to X$ be a proper morphism of schemes which is an isomorphism over $U = X \setminus T$. Then there exists a sequence*

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

where $X_{i+1} \to X_i$ is the blowing up of X_i at a closed point x_i lying above a point of T and a factorization $X_n \to Y \to X$ of the composition.

Proof. By More on Flatness, Lemma 081T there exists a U-admissible blowup $X' \to X$ which dominates $Y \to X$. Hence we may assume there exists an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ such that $\mathcal{O}_X/\mathcal{I}$ is supported on T and such that Y is the blowing up of X in $\mathcal{I}$. By Lemma 0AHH there exists a sequence

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

where $X_{i+1} \to X_i$ is the blowing up of X_i at a closed point x_i lying above a point of T such that $\mathcal{I}\mathcal{O}_{X_n}$ is an invertible ideal sheaf. By the universal property of blowing up (Divisors, Lemma 0806) we find the desired factorization. □